

MINERS : Multilingual Language Models as Semantic Retrievers

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Abstract

Words have been represented in a high-dimensional vector space that encodes their semantic similarities, enabling downstream applications such as retrieving synonyms, antonyms, and relevant contexts. However, despite recent advances in multilingual language models (LMs), the effectiveness of these models' representations in semantic retrieval contexts has not been comprehensively explored. To fill this gap, this paper introduces the MINERS, a benchmark designed to evaluate the ability of multilingual LMs in semantic retrieval tasks, including bitext mining and classification via retrieval-augmented contexts. We create a comprehensive framework to assess the robustness of LMs in retrieving samples across over 200 diverse languages, including extremely low-resource languages in challenging cross-lingual and code-switching settings. Our results demonstrate that by solely retrieving semantically similar embeddings yields performance competitive with state-of-the-art approaches, without requiring any fine-tuning.

1 Introduction

Language models (LMs) play a crucial role in learning natural language representations (Cer et al., 2018; Kenton and Toutanova, 2019; Reimers and Gurevych, 2019; Gao et al., 2021; Feng et al., 2022) and have been successfully applied to various natural language processing (NLP) tasks, such as document retrieval (Yang et al., 2019a; Wang et al., 2023). Existing benchmarks have systematically evaluated LMs to provide empirical assessments of their performance across a range of embedding tasks. Some notable benchmarks include Big-Bench (Srivastava et al., 2023), MTEB (Muennighoff et al., 2023a), SemEval (Cer et al., 2017), and BEIR Benchmark (Thakur et al., 2021). MTEB, in particular, has been established as a comprehensive benchmark for evaluating the

effectiveness of embeddings in downstream NLP applications. However, their analysis of the multilingual space has been limited to bitext mining, without further exploration of how these embeddings can be utilized in other multilingual downstream tasks.

The advancement of multilingual LMs is remarkable, demonstrating impressive capabilities in adapting to new languages through fine-tuning (Conneau and Lample, 2019; Alabi et al., 2022), learning from few-shot samples via in-context learning (ICL) (Lin et al., 2021; Winata et al., 2021b; Tanwar et al., 2023; Cahyawijaya et al., 2024; Biderman et al., 2024), enabling cross-lingual zero-shot transfer (Ruder et al., 2021), and incorporating language-specific adapters (Ansell et al., 2021; Yong et al., 2023). This exploration now includes low-resource and regional languages not part of the pretraining phase, promoting NLP research for underrepresented languages (Adelani et al., 2022; Winata et al., 2022; Song et al., 2023). However, multilingual LMs face two key challenges: (1) the lack of a comprehensive benchmark for evaluating effectiveness in semantic retrieval, and (2) limited understanding of code-switching (CS) texts common in multilingual communities.

Current CS evaluations focus on model fine-tuning benchmarks (Aguilar et al., 2020; Khanuja et al., 2020; Winata et al., 2021a; Zhang et al., 2023), without deeply exploring their potential as multilingual retrievers. Recent studies by Winata et al. (2023a) have primarily focused on semantic similarity using encoder LMs in zero-shot cross-lingual settings but have not explored their application in generative LMs. This gap presents an opportunity to leverage these models as context providers for multilingual generative LMs (Lewis et al., 2020; Bevilacqua et al., 2022).

In this paper, we introduce MINERS, the first benchmark designed to assess the multilingual LMs' ability in semantic retrieval across various

*The work was conducted outside Capital One.

tasks. MINERS evaluates the representation of dense vectors in multiple tasks, including bixtext retrieval, retrieval-based classification, and ICL classification. We have developed MINERS to be a reproducible and reliable benchmark that utilizes high-dimensional multilingual vector representations. Notably, these tasks do not require any fine-tuning. The paper’s contribution can be summarized as follows:

- We introduce **MINERS**, the first comprehensive benchmark designed to systematically evaluate multilingual LMs as semantic retrievers across a vast array of languages. Covering 200+ languages, 11 encoder LMs, and 11 generative LMs, including open-source and commercial APIs. MINERS offers a robust evaluation framework for assessing the effectiveness of LMs in diverse linguistic contexts.
- We show MINERS is highly adaptable and scalable across various models. By consolidating scores from multiple models, MINERS facilitates a comprehensive evaluation of task performance, providing insights into different approaches’ strengths and weaknesses.
- We provide a thorough analysis across different evaluation difficulty levels, including monolingual, cross-lingual, and CS scenarios. We examine performance variations across different numbers of retrieved samples to offer insights into the impact of sample quantity on retrieval effectiveness.
- We compare the time efficiency of retrieval methods with conventional fine-tuning approaches. By demonstrating that retrieval methods require no training and offer a comparable performance of leveraging pre-trained models for semantic retrieval tasks.

2 MINERS BENCHMARK

2.1 Motivation

The MINERS BENCHMARK¹ is introduced as a significant step forward in assessing the capabilities of multilingual LMs in producing high-dimensional representations for semantic retrieval. This benchmark is constructed with three fundamental aspects: **(1) Language Diversity**: The benchmark offers insights into the performance of LMs across a wide

array of languages. It assesses not only the models’ effectiveness in high-resource languages but also their capabilities in low-resource languages from various language families. Additionally, the benchmark includes evaluations of unseen languages to gauge the robustness of the models in predicting languages not encountered during pre-training. CS datasets are also incorporated to simulate realistic scenarios where bilingual or multilingual speakers mix languages, providing a more comprehensive assessment of the models’ capabilities. **(2) Usefulness**: The benchmark includes evaluations across three distinct tasks to systematically measure the performance of multilingual LMs. First, it assesses the models’ ability to retrieve semantically similar parallel data in bixtext retrieval tasks. Second, it uses the retrieved samples for classification, evaluating the models’ accuracy in categorizing text. Third, it employs the retrieved samples as context for generating labels in downstream classification tasks, highlighting the models’ capability to incorporate retrieved information into context-aware classification. Additionally, the benchmark demonstrates the potential of using multiple LMs and APIs together to represent text as an ensemble, further emphasizing their utility. **(3) Efficiency**: The benchmark is crafted with efficiency as a key principle. It is designed to be straightforward and easily extendable, accommodating new datasets to ensure its longevity and continued relevance. Additionally, the benchmark is publicly available, promoting result reproducibility and encouraging collaboration and further research within the field. Importantly, the benchmark does not necessitate any model fine-tuning, as all evaluations are conducted exclusively through model inference, thereby streamlining the assessment process.

2.2 Tasks

Our benchmark evaluates LMs on three tasks: bixtext retrieval, retrieval-based classification, and ICL classification. Figure 1 provides an overview of tasks. We describe the task details as follows:

Bixtext Retrieval This task aims to measure the LM’s ability to retrieve semantically similar samples from parallel datasets. The task is also useful to understand how the model perform when there are language distribution shifts, especially when some words are code-switched. Formally, given a parallel dataset $\mathcal{D}$ with two language L_1 and L_2 , we can have two different datasets $\mathcal{D}_{L_1}$ and $\mathcal{D}_{L_2}$.

¹We release the code to reproduce the benchmark results at <https://github.com/gentaiscool/miners>

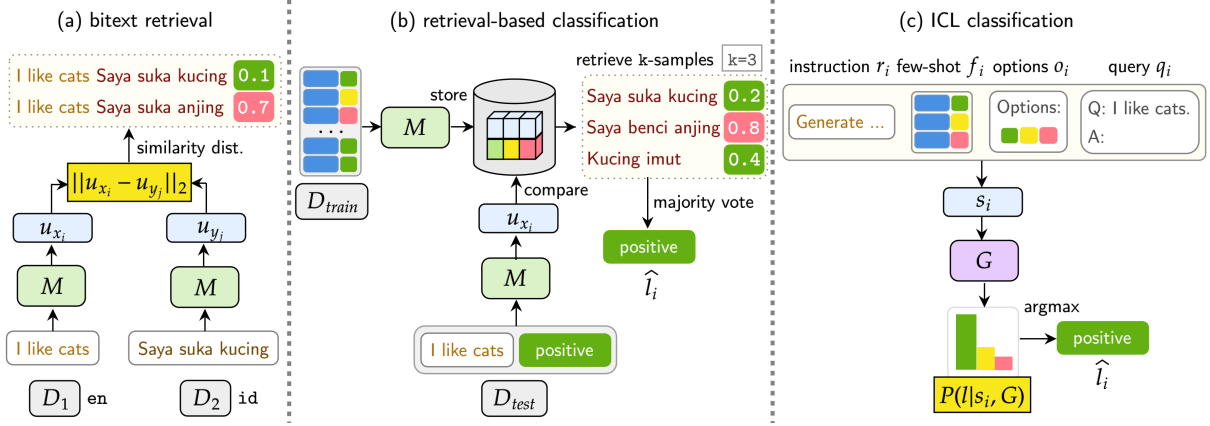


Figure 1: MINERS BENCHMARK tasks. In this example, we compare English (en) and Indonesian (id) texts across three tasks: (a) bitext retrieval, (b) retrieval-based classification, and (c) ICL classification. Light blue cubes represent vector representations of samples from the training dataset $\mathcal{D}_{train}$, generated by $\mathcal{M}$, while green, yellow, and red cubes denote raw text labels. The few-shot samples f_i in task (c) are retrieved in the same manner as in task (b). The English translations of the text in the figure are as follows: "Saya suka kucing" ("I like cats"), "Saya suka anjing" ("I like dogs"), "Saya benci anjing" ("I hate dogs"), and "Kucing imut" ("Cute cats").

For each sample x_i in $\mathcal{D}_{L_1}$, the closest sample $\hat{y}$ is searched through $\mathcal{D}_{L_2}$, by finding the lowest distance score between two samples x_i and y_j . The score $s_{i,j}$ is computed by measuring the Euclidean distance of their high-dimensional vector representation which generated by using an LM $\mathcal{M}$. In this case, euclidean distance is used to compute the score $s_{i,j} = ||\mathbf{u}_{x_i} - \mathbf{u}_{y_j}||_2$, where $\mathbf{u}_{x_i}$ and $\mathbf{u}_{y_j}$ are vector representation of samples x_i and y_j , respectively. We can also use other distance measures, but the difference is minimal.

Retrieval-based Classification This task involves using the retrieved samples' labels from the training set to predict labels in downstream NLP classification tasks. The goal is to assess the usefulness of our retrieved samples and introduce an efficient prediction method by directly searching for similar samples in the training set. Given the retrieved k pairs of training samples with labels $[(y_1, l_1), \dots, (y_k, l_k)]$, a label $\hat{l}$ is selected by majority voting and assigned to the corresponding test sample. Increasing k can enhance performance.

ICL Classification We aim to further utilize the retrieved training samples for natural generation tasks by using them as few-shot context, combined with task-specific instructions and a query. Formally, given a generative LLM G , we input a text sequence $s_i = (r_i; f_i; o_i; q_i)$, which includes a text instruction r_i , few-shot samples $f_i = [(y_1, l_1), \dots, (y_k, l_k)]$, a list of label options o_i , and a query q_i , to generate an output text se-

quence. To generate the prediction, we use one of two methods based on the model's capabilities: (a) computing label probabilities, which offers precise predictions by reducing issues like typos, and (b) directly predicting labels through instructions, which is more efficient as responses match desired labels, eliminating the need to evaluate all options. We use method (a) when we can calculate the log-likelihood of the next token prediction; otherwise, we resort to method (b). For method (a), we compute the probability of each output class, normalize it by the token length, and select the label with the highest probability from the distribution as follows:

$$\hat{l}_i = \arg \max_{l \in L} P(l | s_i, G), \quad (1)$$

where L denotes the number of possible classes. For more details on model inference, please refer to Appendix A.5.

2.3 Settings

We gauge LMs' robustness to various text inputs with three different evaluation settings:

- **Monolingual (Mono):** We measure the individual language performance using the same language as train and test sets.
- **Code-switching (CS):** We measure the performance of mixed language datasets. For bitext retrieval, we find a corresponding CS text translation from a monolingual text, or

Dataset	Lang.	Task	Eval Metric
BUCC (Zweigenbaum et al., 2017, 2018)	5	Bitext Retrieval	F1
MASSIVE (FitzGerald et al., 2023)	51	Intent Classification $\diamond$	Acc.
NollySenti (Shode et al., 2023)	5	Bitext Retrieval	F1
		Sentiment Analysis $\diamond \spadesuit$	Acc.
NusaX (Winata et al., 2023b)	12	Bitext Retrieval	F1
		Sentiment Analysis $\diamond \spadesuit$	F1
NusaT (Cahyawijaya et al., 2023)	12	Bitext Retrieval	F1
SIB-200 (Adelani et al., 2023)	205	Topic Classification $\diamond \spadesuit$	Acc.
Tatoeba (Tiedemann, 2020)	113	Bitext Retrieval	F1
Code-switching			
FIRE 2020 (Chakravarthi et al., 2020; Hegde et al., 2022)	3	Sentiment Analysis $\diamond \spadesuit$	Acc.
LinCE MT (Aguilar et al., 2020)	2	Bitext Retrieval	F1
LinCE SA (Patwa et al., 2020)	2	Sentiment Analysis $\diamond \spadesuit$	Acc.
PHINC (Srivastava and Singh, 2020)	2	Bitext Retrieval	F1

Table 1: Dataset list of MINERS BENCHMARK. The symbols indicate the tasks run on datasets. $\diamond$ Retrieval-based classification task. $\spadesuit$ ICL classification task.

vice versa, and for retrieval-based classification and ICL classification, we take CS texts as input and predict their labels.

- **Cross-lingual (XL):** We measure the performance of multilingual datasets with one language as the source language and the rest as target languages. For detailed information, please refer to Table 7 in the Appendix.
- **Cross-lingual Code-switching (XL CS):** We tackle a more challenging scenario by evaluating CS data within a cross-lingual context.

2.4 Datasets

Table 1 presents 11 datasets: 7 multilingual and 4 CS datasets, covering both parallel and classification types. Parallel datasets are ideal for bitext retrieval due to their aligned multilingual content, enabling bitext mining and machine translation tasks. Classification datasets include intent classification, sentiment analysis, and topic classification, which we evaluate for retrieval-based and ICL classification tasks. For ICL, we construct prompts using a unified English template across all generative language models to ensure simplicity and consistency. Detailed instructions for each task are provided in Tables 17 and 18 in the Appendix.

2.5 Models

Encoder LMs and APIs We use 9 open-source LMs: LaBSE (Feng et al., 2022),

CMLM (Cer et al., 2018), multilingual E5_{BASE}, multilingual E5_{LARGE} (Wang et al., 2024), multilingual MPNet_{BASE}v2 (Song et al., 2020), multilingual MiniLM_{L12-E384} (Wang et al., 2020), Glot-500 (ImaniGooghari et al., 2023), XLM-R_{BASE}, XLM-R_{LARGE} (Conneau and Lample, 2019), and two commercial embedding APIs: Cohere-Embedv3 (embed-multilingual-v3.0) and OpenAI-Embedv3 (text-embedding-3-large).²

Generative LMs We opt for 8 different open-source LMs: (1) BLOOMZ (Muennighoff et al., 2023b), an instruction tuned BLOOM (Le Scao et al., 2023) with three different sizes (560m, 1B, 3B) to further analyze the performance trend when increasing the model size, (2) mT0 3B (x1) (Muennighoff et al., 2023b), an instruction tuned mT5 (Xue et al., 2021), (3) XGLM (Lin et al., 2021) with two different sizes (564m and 2.9B), (4) Aya-23 8B (Aryabumi et al., 2024), (5) Aya-101 13B (Üstün et al., 2024), (6) Gemma 1.1 Instruct (Team et al., 2024), (7) Llama 3 8B Instruct, and (8) Llama 3.1 8B Instruct (Dubey et al., 2024), and three commercial APIs: (1) Command-R, (2) GPT-3.5 Turbo (gpt-3.5-turbo-0125) and (3) GPT-4o (gpt-4o-2024-05-13). All open-source models can be found on Hugging Face. Please check the Appendix on Table 8 for details.

²The APIs were accessed on May 2024.

Model	Bitext Retrieval			Retrieval-based Classification				
	XL	CS	avg.	Mono	XL	CS	XL CS	avg.
Fine-tune (XLM-R _{BASE})	N/A	N/A	N/A	79.55	65.92	62.28	34.64	60.60
LaBSE	83.90	52.03	67.97	73.46	72.73	60.64	41.10	61.98
CMLM	70.77	42.62	56.70	73.05	70.31	59.27	40.88	60.88
E5 _{BASE}	72.26	43.29	57.78	75.08	65.51	61.16	42.73	61.12
E5 _{LARGE}	76.35	49.97	63.16	77.52	71.08	61.91	41.99	63.13
MPNet _{BASE} v2	52.25	25.87	39.06	66.17	59.69	58.33	41.25	56.36
MiniLM _{L12-E384}	24.82	9.90	17.36	63.18	51.16	57.28	39.61	52.81
Glott-500	14.68	16.64	15.66	65.66	51.75	58.11	40.06	53.90
XLM-R _{BASE}	17.79	10.61	14.20	63.62	47.59	58.25	41.02	52.62
XLM-R _{LARGE}	12.45	6.04	9.25	61.76	43.88	57.30	39.47	50.60
Cohere-Embedv3	76.39	53.25	64.82	78.56	<u>72.67</u>	62.12	<u>42.36</u>	63.93
OpenAI-Embedv3	69.02	68.73	68.88	73.97	67.13	62.77	40.50	61.09
DistFuse (2) [†]	84.72	56.47	70.60	78.34	70.87	62.13	40.73	63.02
DistFuse (3) [†]	<u>83.28</u>	<u>56.83</u>	<u>70.06</u>	<u>78.80</u>	70.19	<u>62.31</u>	41.77	<u>63.27</u>

Table 2: Results for bitext retrieval task ($k = 1$) and retrieval-based classification ($k = 10$). **Mono**, **XL** and **CS** denote monolingual, cross-lingual and code-switching, respectively. **Bold** and underlined numbers present the best and second-best models. [†] For DistFuse (2), we use $\alpha = 1, \beta = 3$ and for DistFuse (3), we use $\alpha = 1, \beta = 2, \gamma = 3$. The reported weights represent the best-performing configurations identified during our tuning process.

Ensemble Models To enhance scalability and effectiveness, we can use multiple models with DistFuse (Winata et al., 2023a) to improve retrieval results. DistFuse combines models by calculating distance scores of label distributions and merging them through a linear combination. We report two DistFuse settings for bitext retrieval and retrieval-based classification tasks:

- **DistFuse (2)** utilizes two models: LaBSE and E5_{LARGE};
- **DistFuse (3)** utilizes three models: LaBSE, E5_{LARGE}, and Cohere-Embedv3.

To maintain conciseness, we denote the weights assigned to distances computed by LaBSE, E5_{LARGE}, and Cohere-Embedv3 as α, β, γ , respectively.

3 Results

3.1 Bitext Retrieval

Table 2 highlights DistFuse (2) and OpenAI-Embedv3-large as top performers in XS and CS tasks, respectively, with LaBSE ranking highest among open-source models. DistFuse (2) demonstrates superior performance across various settings. While XLM-R and Glot-500 struggle in bitext retrieval, they perform better in retrieval-based classification. Most models face challenges in CS tasks

for both bitext retrieval and retrieval-based classification, where APIs generally perform slightly better. OpenAI-Embedv3 outperforms Cohere-Embedv3 on CS datasets. The specifics of CS training data remain unclear, potentially explaining the APIs’ edge over open-source models. Combining model scores significantly boosts performance, with up to a 2.63% improvement in bitext retrieval over LaBSE and a 1.72% improvement over OpenAI-Embedv3. Similar gains are observed in retrieval-based classification, where the leading DistFuse model, though slightly behind Cohere-Embedv3, notably surpasses OpenAI-Embedv3.

3.2 Retrieval-based Classification Results

Table 2 illustrates that the Cohere-Embedv3 API outperforms all models by an average of 1.95%, with LaBSE closely behind at 1.15%. XLM-R and Glot-500 excel in classification tasks. Despite this, they lag behind models trained with contrastive learning or alignment objectives like LaBSE, CMLM, or E5 models, emphasizing the significance of text alignment in NLP tasks. Merging model scores notably boosts prediction accuracy, especially in Mono and XL settings. However, performance in CS and XL CS settings remains lower compared to API models. Additionally, our

Model	Zero-shot ICL					One-shot ICL				
	Mono	XL	CS	XL CS	avg.	Mono	XL	CS	XL CS	avg.
BLOOMZ 560M	45.88	43.36	35.83	12.09	34.29	72.37	71.98	54.25	36.35	58.74
BLOOMZ 1.7B	54.10	52.86	35.70	11.80	38.62	71.38	70.65	58.04	38.50	59.64
BLOOMZ 3B	53.20	51.78	36.32	9.50	37.70	74.08	73.19	<u>57.44</u>	39.09	60.95
mT0 3B	53.29	53.64	40.11	42.51	47.39	59.02	57.86	<u>46.66</u>	42.36	51.48
XGLM 564m	39.25	37.19	29.92	10.46	29.21	37.26	40.12	22.64	12.83	28.21
XGLM 2.9B	42.41	40.16	34.71	10.39	31.92	42.57	48.76	27.45	10.39	32.29
Aya-23 8B	39.88	36.88	53.72	43.18	43.42	63.66	63.53	53.12	38.50	54.70
Aya-101 13B	<u>78.65</u>	<u>77.72</u>	42.29	26.26	56.23	<u>81.00</u>	<u>80.20</u>	50.90	36.20	<u>62.08</u>
Gemma 1.1 7B Instruct	55.51	53.36	51.62	37.24	49.43	65.82	64.49	53.12	35.68	54.78
Llama 3 8B Instruct	62.40	60.41	52.72	36.05	52.90	74.85	69.61	54.12	35.68	58.57
Llama 3.1 8B Instruct	60.59	58.86	47.99	26.56	48.50	72.68	59.00	54.11	35.16	55.24
Command-R	47.98	46.02	54.84	44.44	48.32	58.36	56.89	56.84	41.99	53.52
GPT-3.5 Turbo	67.10	65.13	<u>54.32</u>	<u>45.18</u>	<u>57.93</u>	71.01	71.56	57.13	<u>42.73</u>	60.61
GPT-4o	79.92	79.15	53.48	53.04	66.40	82.24	80.95	57.14	49.26	67.40

Table 3: Results on ICL classification with E5_{LARGE} retriever. **Bold** and underlined numbers present the best and second-best models.

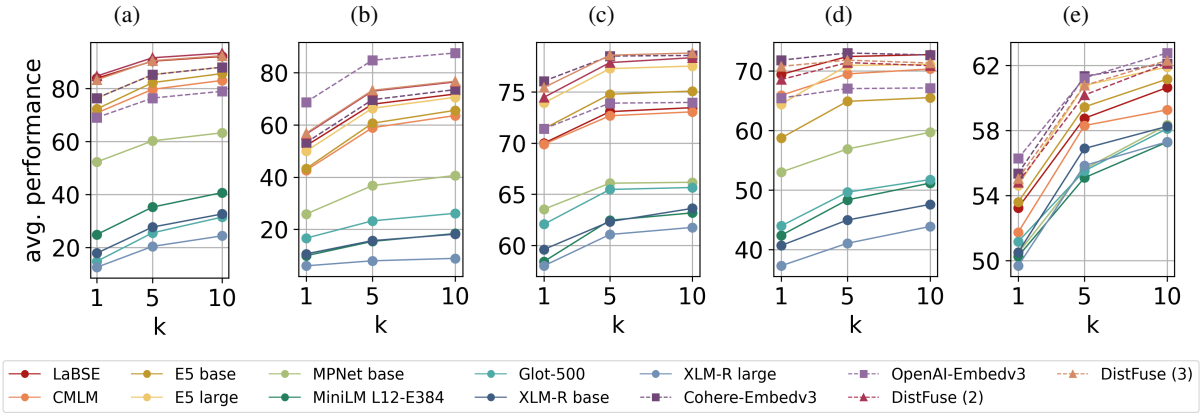


Figure 2: Results with different $k = [1, 5, 10]$ on bitext retrieval: (a) cross-lingual and (b) code-switching, retrieval-based classification: (c) monolingual, (d) cross-lingual, and (e) code-switching.

model outperforms fine-tuned models, requiring no fine-tuning in XL and CS tasks.

3.3 ICL Classification Results

Based on Table 3, we present the ICL classification results using E5_{LARGE} as the retriever. Please see Appendix Table 16 for results from alternate retrievers. The inclusion of few-shot context significantly improves the generative LM’s precision in predicting class labels, leading to enhancements. There is a positive scaling law with increased model size in the one-shot setup. For instance, using a model with 6× more parameters (BLOOMZ 3B) boosts performance by 2.21% compared to the top BLOOMZ 560m model. However, performance decreases for CS and XL CS tasks with increasing

complexity. Despite focusing on English, Llama 3 and Llama 3.1 models generally outperform multilingual open-source models like BLOOMZ, mT0, XGLM, and Aya-23. BLOOMZ excels in the one-shot scenario, outperforming both Llama models. Notably, mT0 outperforms XGLM and Aya-23 in zero-shot settings, despite Aya-23’s larger size. Aya-101 is the top open-source LM in both zero-shot and one-shot tasks, bridging the gap with commercial APIs like GPT-4o. Commercial generative LM APIs, such as GPT-3.5 Turbo and GPT-4o outperform all other models, particularly in CS and XL CS contexts. However, their superior performance may be attributed to prior exposure to these datasets, though this aspect remains unclear.

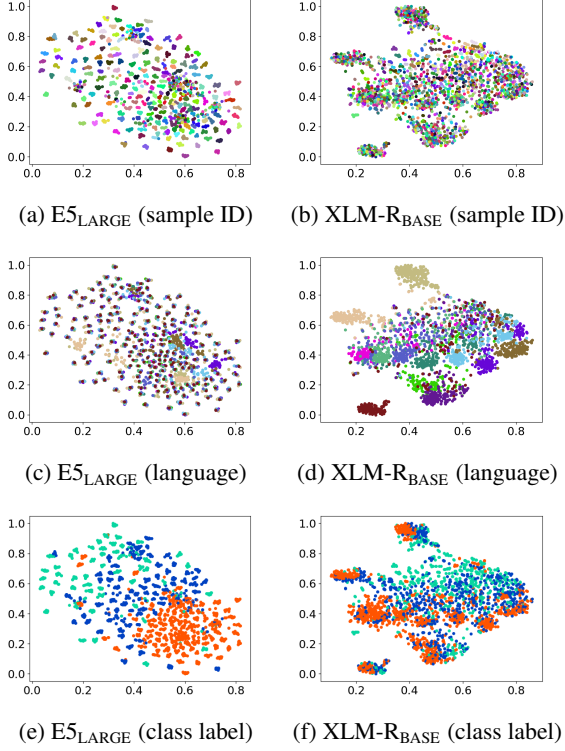


Figure 3: t-SNE representation of 200 randomly training samples from the NusaX dataset. The color on the figures show the sample ID for (a) and (b), language for (c) and (d), and class for (e) and (f).

3.4 Performance Dynamics Over k

Figure 2 shows a consistent positive trend as the retrieved sample size increases for both bitext retrieval and retrieval-based classification tasks. This indicates that model performance improves with more retrieved samples. In bitext retrieval, a larger k provides a richer set of bilingual text pairs, enhancing retrieval. Similarly, in retrieval-based classification, a larger k offers more contextual examples, leading to more precise label predictions through majority voting.

4 Further Analysis

4.1 Model Representation

Figure 3 shows 2D scatter plots of the vector representation generated using t-SNE (Van der Maaten and Hinton, 2008). We take 200 random training samples from the NusaX dataset, reduce the high-dimensional vectors into 2D and color the scatter plots in three ways. (1) **By sample ID**. We assign the same color for parallel samples. (2) **By language**. We assign a color for each language. (3) **By class label**. We assign a color for each class label. We observe that the E5_LARGE model forms

Fine-tune	
(1) Train	$n_{\text{epoch}} \times (\mathcal{D}_{\text{train}} \times (f_{\mathcal{M}} + b_{\mathcal{M}}) + \mathcal{D}_{\text{dev}} \times f_{\mathcal{M}})$
(2) Evaluate	$ \mathcal{D}_{\text{test}} \times f_{\mathcal{M}}$
Retrieval-based Classification	
(1) Generate vectors	$(\mathcal{D}_{\text{train}} + \mathcal{D}_{\text{test}}) \times f_{\mathcal{M}}$
(2) Retrieve samples	$ \mathcal{D}_{\text{train}} \times \mathcal{D}_{\text{test}} \times (n_{\text{dim}} \times (p_+ + p_- + p_{\text{sq}}) + p_{\sqrt{\cdot}})$
ICL Classification	
(1) Generate vectors	$(\mathcal{D}_{\text{train}} + \mathcal{D}_{\text{test}}) \times f_{\mathcal{M}}$
(2) Retrieve samples	$ \mathcal{D}_{\text{train}} \times \mathcal{D}_{\text{test}} \times (n_{\text{dim}} \times (p_+ + p_- + p_{\text{sq}}) + p_{\sqrt{\cdot}})$
(3a) Generate probability	$ \mathcal{D}_{\text{test}} \times f_{\mathcal{G}} \times L \times \bar{L} $
(3b) Generate responses	$ \mathcal{D}_{\text{test}} \times f_{\mathcal{G}} \times L $

Table 4: FLOPs computation formulae. Here, n_{epoch} and n_{dim} denote the number of epochs and vector dimension, respectively. $f_{\mathcal{M}}$ and $b_{\mathcal{M}}$ represent the forward and backward FLOPs of model $\mathcal{M}$, respectively. $f_{\mathcal{G}}$ denotes the forward FLOPs of model $\mathcal{G}$. The symbols p_+ , p_- , p_{sq} , and $p_{\sqrt{\cdot}}$ indicate the FLOPs required to perform the operations of addition, subtraction, squaring, and square root, respectively. Additionally, $|L|$ and $|\bar{L}|$ denote the number of labels and the average sequence length of the labels, respectively. The variables $|\mathcal{D}_{\text{train}}|$, $|\mathcal{D}_{\text{dev}}|$, and $|\mathcal{D}_{\text{test}}|$ represent the sizes of the train, development, and test data splits, respectively.

small, color-coded clusters based on sample ID, indicating its proficiency in aligning text across different languages. In contrast, the XLM-R_BASE model forms larger clusters where samples of the same language group closely together, suggesting it is more effective at identifying same-language data, even for unseen languages in NusaX. However, XLM-R_BASE displays a sparse distribution when classifying samples by sample ID, aligning with our bitext retrieval task results. Both models effectively distinguish label classes, with E5_LARGE achieving better color separation than XLM-R_BASE, as shown in Figures 3 (e) and (f). Similar findings are observed for other models. For more details, refer to Appendix B.1.

4.2 Samples Relevance

Figure 4 illustrates the performance dynamics of BLOOMZ models on the NusaX dataset when retrieving samples from various training data percentiles. Lower percentiles correspond to samples that are more semantically similar to the query. The results indicate that as the percentile decreases, performance improves consistently across all three models. This trend highlights the critical importance of retrieving highly relevant samples for in-context learning (ICL) tasks. By focusing on semantically aligned samples, the models are able to enhance the contextual understanding, which in turn leads to more accurate and reliable predictions. These findings highlight the potential benefits of

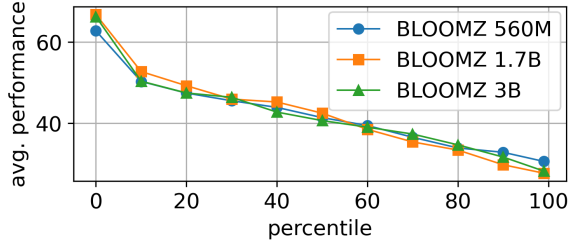


Figure 4: ICL performance dynamics of BLOOMZ models on the NusaX dataset using context retrieved from various percentiles with E5_{LARGE}. Lower percentiles correspond to more semantically relevant samples.

optimizing sample retrieval strategies to improve model performance in various ICL applications.

4.3 Compute Efficiency

We aim to measure the theoretical time complexity by evaluating computation in terms of FLOPs (Floating Point Operations), irrespective of the machine configuration. Table 4 details the components contributing to this calculation. The time complexity for fine-tuning a model scales with the number of training epochs, with more epochs significantly increasing complexity. The backward pass FLOPs, which are substantially higher than forward pass FLOPs, are a major factor. Retrieval-based classification is much more efficient, relying primarily on generating vector representations through forward passes. The retrieval process itself is efficient, with complexity influenced mainly by the sizes of the training and test datasets—factors typically smaller than the computational demands of fine-tuning. In contrast, ICL classification incurs higher inference costs due to the increased forward FLOPs of generative models. With very large LMs, the inference cost can even exceed that of fine-tuning. However, as the training data size increases, the complexity of fine-tuning eventually surpasses ICL model inference. For ICL classification, we have two methods: (a) computing label probabilities, which offers precise predictions, and (b) directly predicting labels through instructions, which is more efficient as responses match desired labels, eliminating the need to evaluate all options. While direct prediction may generate extraneous tokens, this can be mitigated with additional instructions to output only the label.

4.4 Bitext Retrieval is Unsymmetrical

We evaluate the bitext retrieval performance with different source and target language(s) directions. Based on the results presented in Table 5, it is evident that the bitext retrieval performance is asym-

Model	x→eng		eng→x	
	BUCC	Tatoeba	BUCC	Tatoeba
LaBSE	98.77	83.76	98.93	80.31
E5 _{LARGE}	98.66	75.73	98.90	75.98
Glott-500	17.90	10.58	16.39	14.07
XL _M -R _{BASE}	39.70	12.62	24.70	8.61
XL _M -R _{LARGE}	26.51	6.57	11.95	3.30
Cohere-Embedv3	98.76	74.66	98.89	76.43

Table 5: Bitext retrieval F1@1 performance on two different source-to-target language(s) directions. **Bold** and underlined numbers present the best and second-best models.

metrical. Specifically, we observe that using non-English data to retrieve English data tends to be more effective than the reverse scenario.

5 Related Work

Dense Retrieval via LM Dense retrieval has marked a significant advancement in information retrieval, enabling rapid sample searches across vast document collections. Research has focused on training objectives and architectures that produce similarity scores between text samples. Reimers and Gurevych (2019) introduce a Siamese network architecture trained with contrastive learning, enhancing retrieval by enabling vector representation comparison using similarity measures, applied to BERT (Kenton and Toutanova, 2019). Efforts to improve alignment include incorporating annotated pairs from natural language inference datasets using SimCSE loss (Gao et al., 2021). Furthermore, Feng et al. (2022) propose combining monolingual and translation alignment losses to enhance performance, such as masked language modeling (MLM) (Devlin et al., 2019) and translation language modeling (TLM) objectives (Conneau and Lample, 2019), dual encoder translation ranking (Guo et al., 2018), and additive margin softmax (Yang et al., 2019b). Khattab and Zaharia (2020) introduce a late interaction paradigm, comparing embedding representations via vector similarity indexes for relevance estimation in ranking tasks. Wang et al. (2024) further innovate by using in-batch negatives to leverage weakly supervised data from diverse, heterogeneous sources.

Semantic Retrieval for NLP Tasks Retrieving labels using semantic retrieval has proven beneficial for classification. Bari et al. (2021) enhance accuracy with cross-lingual few-shot nearest neigh-

bor adaptation. Winata et al. (2023a) predict test data labels efficiently using English training data without prior adaptation via ICL. Li et al. (2023) introduce a ranking framework to retrieve high-quality demonstrations for various tasks. Building on these methods, we adopt a straightforward and efficient retrieval approach similar to Winata et al. (2023a), supporting multiple retrieval models for open-source tools and APIs. We extend this approach to the ICL setting, enhancing its utility and accessibility across diverse scenarios.

6 Conclusion

This paper introduces MINERS, a benchmark for evaluating the efficacy of multilingual LMs in semantic retrieval tasks, including bitext retrieval and classification through semantic search and retrieval-augmented contexts. Our framework rigorously assesses LMs’ robustness in retrieving samples from over 200 languages. Empirical results demonstrate that our method, which focuses on retrieving semantically similar vector representations, achieves performance comparable to state-of-the-art fine-tuned approaches, without requiring fine-tuning across multiple datasets and languages. We also explore the mechanisms behind these representations, offering insights to improve the efficiency and accuracy of label retrieval methods. Our research aims to pave the way for future exploration and optimization in semantic retrieval and classification, ultimately contributing to more robust and adaptable NLP systems.

Limitations

We have identified potential avenues for enhancing the performance of the ICL classification task through the application of ensemble techniques such as DistFuse and using the target language prompts instead of English. Additionally, while we have primarily focused on evaluating the BLOOMZ, mT0, XGLM, Gemma, Llama 3, Llama 3.1, Aya-23, Aya-101, Command-R, GPT-3.5 Turbo, and GPT-4o models within the benchmark, we acknowledge that there may be other models that could also yield promising results. These aspects represent areas for future exploration and expansion of our research efforts. Due to resource limitations and simplicity, we only test a single prompt template. Running with various prompts could yield different results, but we defer this exploration to future research.

In the future, we plan to explore deeper into the capabilities of ensemble techniques like DistFuse to further improve the performance of the ICL classification task. By combining the strengths of multiple models, we aim to enhance the robustness and accuracy of our classification outcomes, ultimately achieving better results in real-world applications. Furthermore, our current evaluation has been limited to a select few models and datasets as part of our initial assessment phase. However, we recognize the importance of conducting a more comprehensive evaluation by considering a wider range of models and datasets. This will allow us to gain a more comprehensive understanding of the strengths and weaknesses of different approaches, enabling us to make more informed decisions about model selection and optimization strategies.

Ethical Considerations

Our research aims to evaluate LMs in the context of multilingual semantic retrieval, a field with significant implications for diverse multilingual communities. We strive to ensure that our evaluation is conducted with the utmost transparency and fairness.

Acknowledgments

We would like to extend our gratitude to Zheng-Xin Yong for his insightful comments and suggestions on this work. We also appreciate the insightful reviews from the anonymous reviewers, which have contributed to improving the quality of this work. David Adelani is supported by Canada CIFAR AI Chair program.

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A Experimental Details

A.1 Baselines

For the task-specific evaluation, we include the following baseline models for comparison:

SOTA We report the state-of-the-art (SOTA) from the existing literature as follows:

- **Bitext Retrieval:** BUCC (Wang et al., 2024) and Tatoeba (Wang et al., 2024).
- **Classification:** MASSIVE (FitzGerald et al., 2023), NollySenti (Shode et al., 2023), NusaX (Winata et al., 2023b, monolingual) (Winata et al., 2023a, cross-lingual), and SIB-200 (Adelani et al., 2023). We use the validation split on Accuracy for LinCE SA, but to the best of our knowledge, there is no comparable result in the literature. We make a small modification to FIRE 2020 labels, thus there are no comparable results in the literature.

Classification Baselines We report the following baselines for classification tasks:

- **Random:** In this baseline, prediction labels are sampled randomly from a uniform distribution. This approach ensures that each label has an equal probability of being selected, regardless of its true distribution within the dataset. It serves as a baseline to compare the effectiveness of more sophisticated methods.
- **Majority:** In this baseline, prediction labels are selected by taking the majority class for all instances. By always predicting the most frequent class observed in the training data,

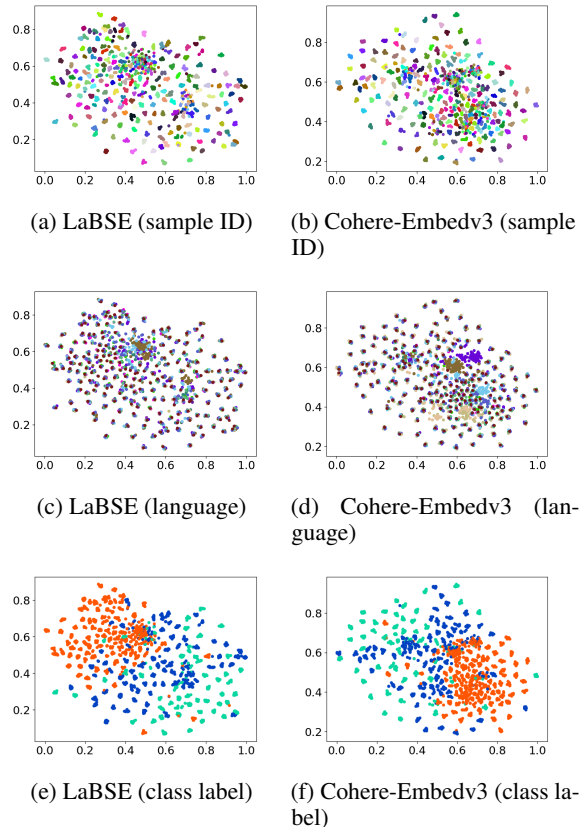


Figure 5: t-SNE representation of 200 random samples from the NusaX dataset. The color on the figures show the sample ID for (a) and (b), language for (c) and (d), and class for (e) and (f).

this method provides a simple yet effective baseline, especially in datasets with class imbalance. It helps to highlight the performance of models in recognizing and classifying less frequent classes.

- **Fine-tune (XLM-R_{BASE}):** We fine-tune a XLM-R_{BASE} model using the training split of the dataset. After fine-tuning, the model is evaluated on the test data split of the same dataset to assess its performance.

A.2 Datasets

A.2.1 Preprocessing

To enhance the data clarity for LMs and improve their predictive performance, we apply preprocessing steps to the following two datasets:

- **FIRE 2020:** We modify several non-standard labels to a single label for sentiment analysis. We map “Mixed_feeling” into “Mixed”, and map “not-malayalam”, “non-tamil”, and “unknown_state” into “Unknown”.

- **MASSIVE**: We replace the underscore character with a space character from the labels.

A.2.2 Statistics

Table 6 displays the dataset statistics for each dataset split. In the case of NollySenti in bitext-retrieval, the English data predominates over other languages, prompting us to consider an equal number of data points for all languages. As for LinCE SA, since we did not utilize the test set, the statistics for this particular dataset are not reported.

A.3 Languages Under Study

Table 7 presents a comprehensive list of source and target language pairs used in our cross-lingual experiments. The datasets apply different language code standards. To ensure consistency and uphold the integrity of the original datasets, we have reported the language codes exactly as they appear in the respective sources.

A.4 LM Sources

We extensively utilize a range of open-source encoder and generative LMs from the Hugging Face repository to ensure our evaluations are comprehensive and transparent. The models we employ are detailed in Table 8, showcasing the diversity in architectures and training objectives. These open-source models provide a solid foundation for our evaluations, allowing us to benchmark against widely accepted standards in the NLP community. For commercial models, we leverage state-of-the-art APIs to access robust and high-performance LMs. Specifically, we use the OpenAI API to retrieve generation responses from GPT-3.5 Turbo and GPT-4. Additionally, we utilize Cohere’s Embed API to incorporate the Cohere-Embedv3 model.

A.5 LM Inference

We run our model inference on an A100 40G GPU, utilizing 8-bit quantization (Dettmers et al., 2022) to optimize memory usage and speed up inference. Our experiments investigate the impact of varying the number of retrieved samples $k \in [1, 5, 10]$ to understand how retrieval quality and classification performance change with the number of instances. These samples are used for both bitext retrieval and retrieval-based classification tasks. For the ICL classification task, we evaluate our model in both zero-shot and one-shot scenarios using two methods: (1) predicting the label distribution by

computing the next token probability, and (2) generating the response directly. For BLOOMZ, Aya, and XGLM models, we use the first method since we have access to the next token prediction logits. For Llama 3, Gemma, and mT0 models, obtaining these logits is less straightforward. Specifically, the presence of numerous special tokens in Llama 3 complicates logit calculation, so we opt for the second method, which leverages the model’s strong capability to generate exact labels by following instructions. Similarly, for GPT-3.5 Turbo and GPT-4o models, we adopt the second method because we do not have direct access to the logits for all possible classes. These models excel in instruction following, making direct response generation a practical and effective approach.

A.6 Hyper-parameters

To ensure fair and consistent evaluations across our models, we employ a set of specific hyper-parameters during the inference stage, as detailed in Table 10. These hyper-parameters have been carefully chosen to standardize the evaluation process and ensure that our comparisons are both meaningful and reliable. For our fine-tuning baselines, we adopt a different set of hyper-parameters, which are listed comprehensively in Table 9. These parameters are optimized to enhance the model’s performance during the fine-tuning phase. Moreover, to streamline the fine-tuning process, we have decided not to incorporate any warmup steps. The linear scheduler has been chosen for its simplicity and effectiveness.

B Extended Analysis

B.1 LM Representation Visualization

In Figure 5, we present the t-SNE 2D visualization of a subset of 200 randomly selected samples from the NusaX dataset. The visualization showcases how the LaBSE and Cohere-Embedv3 models effectively align samples originating from various languages in a meaningful and interpretable manner. Notably, both models exhibit a high level of proficiency in grouping the samples based on their class labels, indicating robust performance in semantic alignment tasks. This finding is consistent with the behavior observed in models that have been trained using contrastive learning methods, such as the E5 models. The ability of these models to accurately capture semantic relationships across multilingual data highlights their effectiveness in

Dataset	lang	# Train	# Valid	# Test	Source	License
BUCC	all	N/A	N/A	35k	https://huggingface.co/datasets/mteb/bucc-bitext-mining	CC-BY-SA
MASSIVE	all	587k	104k	152k	https://huggingface.co/datasets/AmazonScience/massive	CC-BY 4.0
NollySenti	en	1,302	100	500	https://github.com/IyanuSh/NollySenti/tree/main	CC-BY 4.0
	yo	900	100	500		
	ha/ig/pcm	410	100	500		
NusaX	each lang.	500	100	400	https://huggingface.co/datasets/indonlp/NusaX-senti/viewer/eng/train	CC-BY-SA 4.0
NusaT	btb/bew/jav/	6.6k	849	2k	https://huggingface.co/datasets/indonlp/nusatranslation_mt	Apache 2.0
	mad/mak/min/sun	6.6k	849	2k		
	abs/bhp/mui/rej	1k	174	400		
Code-switching						
FIRE 2020	malayalam	4,851	541	1,348	https://dravidian-codemix.github.io/2020/	N/A
	tamil	11,335	1,260	3,149		
LinCE MT	eng-hinglish	8,060	942	N/A	https://ritual.uh.edu/lince/	Research Only
LinCE SA	eng-spa	12,002	2,998	N/A	https://huggingface.co/datasets/lince-benchmark/lince	Research Only
PHINC	N/A	N/A	27,477		https://huggingface.co/datasets/veezbo/phinc	CC-BY 4.0

Table 6: Dataset statistics.

Dataset	Source Language	Target Language(s)
BUCC	en	de, fr, zh
FIRE 2020	tamil	malayalam
MASSIVE	en	af, am, ar, az, bn, cy, da, de, el, es, fa, fi, fr, he, hi, hu, hy, id, is, it, ja, jv, ka, km, kn, ko, lv, ml, mn, ms, my, nb, nl, pl, pt, ro, ru, sl, sq, sv, sw, ta, te, th, tl, tr, ur, vi, zh-CN, zh-TW
NollySenti	en	ha, ig, pcm, yo
NusaX	eng	ace, ban, bbc, bjn, bug, ind, jav, mad, min, nij, sun
SIB-200	eng_Latn	ace_Arab, ace_Latn, acm_Arab, acq_Arab, aeb_Arab, afr_Latn, ajp_Arab, aka_Latn, als_Latn, amh_Ethi, apc_Arab, arb_Arab, arb_Latn, ars_Arab, ary_Arab, arz_Arab, asm_Beng, ast_Latn, awa_Deva, ayr_Latn, azb_Arab, azj_Latn, bak_Cyrl, bam_Latn, ban_Latn, bel_Cyrl, bem_Latn, ben_Beng, bho_Deva, bjn_Arab, bod_Tibt, bos_Latn, bug_Latn, bul_Cyrl, cat_Latn, ceb_Latn, ces_Latn, cjk_Latn, ckb_Arab, crh_Latn, cym_Latn, dan_Latn, deu_Latn, dik_Latn, dyu_Latn, dzo_Tibt, ell_Grek, epo_Latn, est_Latn, eus_Latn, ewe_Latn, fao_Latn, fij_Latn, fin_Latn, fon_Latn, fra_Latn, fur_Latn, fuv_Latn, gaz_Latn, gla_Latn, gle_Latn, glg_Latn, grm_Latn, guj_Gujr, hat_Latn, hau_Latn, heb_Hebr, hin_Deva, hne_Deva, hrv_Latn, hun_Latn, hye_Armen, ibo_Latn, ilo_Latn, ind_Latn, isl_Latn, ita_Latn, jav_Latn, jpn_Jpan, kab_Latn, kac_Latn, kam_Latn, kan_Knda, kas_Arab, kas_Deva, kat_Geor, kaz_Cyrl, kbp_Latn, kea_Latn, khk_Cyrl, khm_Khmr, kik_Latn, kin_Latn, kir_Cyrl, kmb_Latn, kmr_Latn, knc_Arab, knc_Latn, kon_Latn, kor_Hang, lao_Lao, lij_Latn, lim_Latn, lin_Latn, lit_Latn, lmo_Latn, ltg_Latn, ltz_Latn, lua_Latn, lug_Latn, luo_Latn, lus_Latn, lvs_Latn, mag_Deva, mai_Deva, mal_Mlym, mar_Deva, min_Arab, min_Latn, mkd_Cyrl, mlt_Latn, mni_Beng, mos_Latn, mri_Latn, mya_Mymr, nld_Latn, nno_Latn, nob_Latn, npi_Deva, nqo_Nkoo, nso_Latn, nus_Latn, nya_Latn, oci_Latn, ory_Orya, pag_Latn, pan_Guru, pap_Latn, pbt_Arab, pes_Arab, plt_Latn, pol_Latn, por_Latn, prs_Arab, quy_Latn, ron_Latn, run_Latn, rus_Cyrl, sag_Latn, san_Deva, sat_Olck, scn_Latn, shn_Mymr, sin_Sinh, slk_Latn, slv_Latn, smo_Latn, sna_Latn, snd_Arab, som_Latn, sot_Latn, spa_Latn, srd_Latn, srp_Cyrl, ssw_Latn, sun_Latn, swe_Latn, swb_Latn, szl_Latn, tam_Taml, taq_Latn, taq_Tfng, tat_Cyrl, tel_Telu, tgk_Cyrl, tgl_Latn, tha_Thai, tir_Ethi, tpi_Latn, tsn_Latn, tso_Latn, tuk_Latn, tum_Latn, tur_Latn, twi_Latn, tzm_Tfng, uig_Arab, ukr_Cyrl, umb_Latn, urd_Arab, uzn_Latn, vec_Latn, vie_Latn, war_Latn, wol_Latn, xho_Latn, ydd_Hebr, yor_Latn, yue_Hant, zho_Hans, zho_Hant, zsm_Latn, zul_Latn
Tatoeba	eng	afr, amh, ang, ara, arq, arz, ast, awa, aze, bel, ben, ber, bos, bre, bul, cat, cbk, ceb, ces, cha, cmn, cor, csb, cym, dan, deu, dsb, dtp, ell, epo, est, eus, fao, fin, fra, fry, gla, gle, glg, gsw, heb, hin, hrv, hsb, hun, hye, ido, ile, ina, ind, isl, ita, jav, jpn, kab, kat, kaz, khm, kor, kur, kzj, lat, lfn, lit, lvs, mal, mar, max, mhr, mkd, mon, nds, nld, nno, nob, nov, oci, orv, pam, pes, pms, pol, por, ron, rus, slk, slv, spa, sqi, srp, swe, swg, swb, tam, tat, tel, tgl, tha, tuk, tur, tzl, uig, ukr, urd, uzb, vie, war, wuu, xho, yid, yue, zsm

Table 7: List of source and target languages for all datasets in the cross-lingual setting. Each dataset employs a different language code standard, and we have reported them as used.

handling diverse linguistic contexts and tasks.

B.2 Retrieved Samples

We conduct a detailed comparison of the retrieved samples to assess their quality in terms of semantic relevance to the query. Table 11 presents a comparative analysis between the retrieved samples from E5_{LARGE} and XLM-R_{BASE}. Moreover, Table 12 showcases the retrieved samples from LaBSE. Our evaluation reveals that the samples retrieved from E5_{LARGE} and LaBSE predominantly contain cor-

rect labels, with four out of five labels being accurate. In contrast, the samples retrieved by XLM-R_{BASE} exhibit a lower accuracy rate, with only two out of five labels being correct. This analysis underscores the varying performance in sample quality and label accuracy across the different models, emphasizing the significance of retrieval quality in downstream tasks.

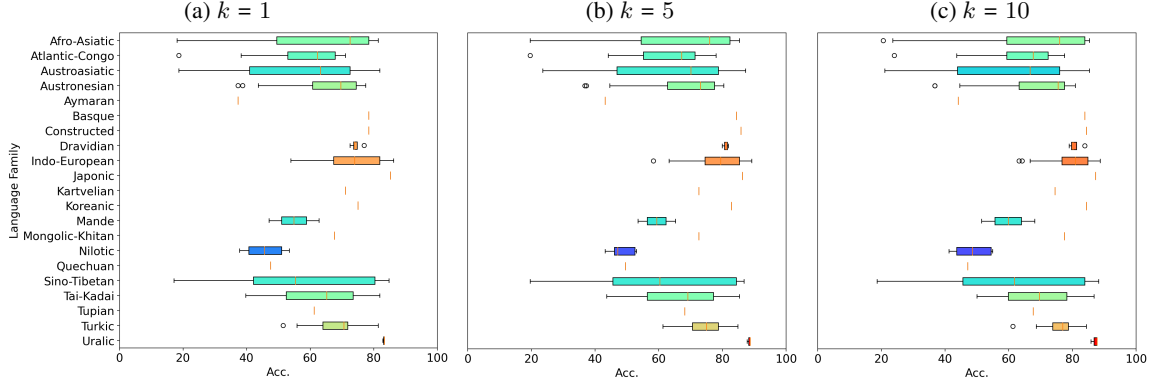


Figure 6: Results for the retrieval-based classification task on the SIB-200 dataset, using k values of [1, 5, 10], across various language families.

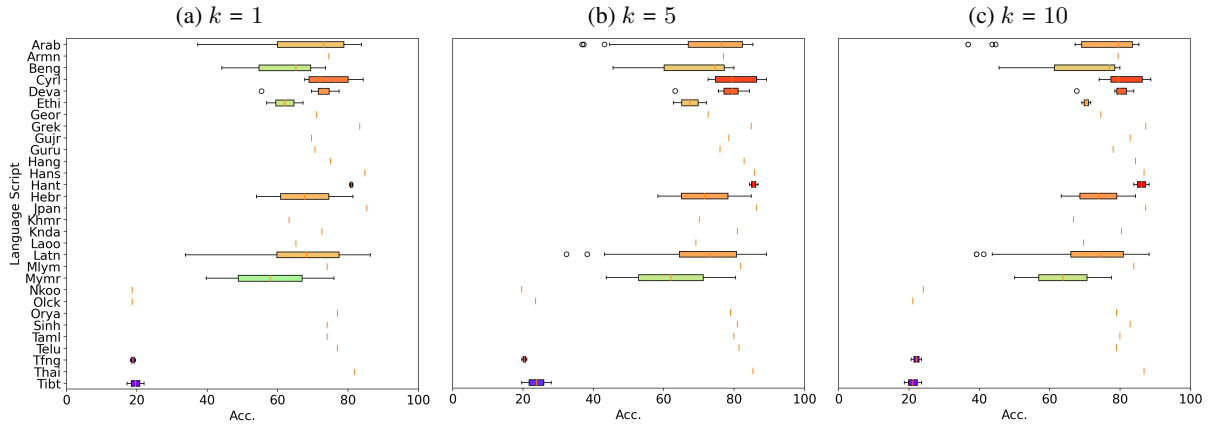


Figure 7: Results for the retrieval-based classification task on the SIB-200 dataset, using k values of [1, 5, 10], across various language scripts.

Model	Hugging Face Model
LaBSE	sentence-transformers/LaBSE
CMLM	sentence-transformers/use-cmlm-multilingual
E5 _{BASE}	intfloat/multilingual-e5-base
E5 _{LARGE}	intfloat/multilingual-e5-large
MPNet _{BASE} V2	sentence-transformers/paraphrase-multilingual-mpnet-base-v2
MiniLM _{L12-E384}	microsoft/Multilingual-MiniLM-L12-H384
Glott-500	cis-lmu/glott500-base
XML-R _{BASE}	FacebookAI/xlm-roberta-base
XML-R _{LARGE}	FacebookAI/xlm-roberta-large
Aya-23 8B	CohereForAI/aya-23-8B
Llama 3 8B Instruct	meta-llama/Meta-Llama-3-8B-Instruct
Llama 3.1 8B Instruct	meta-llama/Meta-Llama-3.1-8B-Instruct
BLOOMZ 560m	bigscience/bloomz-560m
BLOOMZ 1.7B	bigscience/bloomz-17b
BLOOMZ 3B	bigscience/bloomz-3b
mT0 3B	bigscience/mt0-xl
XGLM 564m	facebook/xglm-564M
XGLM 2.9B	facebook/xglm-2.9B

Table 8: Hugging Face models.

C Detailed Results

C.1 Bitext Retrieval Results

Table 13 presents the complete empirical results for each dataset and model in the bitext retrieval task. Generally, there is a positive trend in model performance as the number of k samples increases.

C.2 Retrieval-based Classification Results

Table 14 presents the complete results for the retrieval-based classification task in both Mono and CS settings. Table 15 provides the full results for the XS and XS CS settings. Figure 6 presents the performance results across various language families on the SIB-200 dataset for different values of k . Notably, Indo-European languages consistently achieve the highest accuracies. In contrast, Afro-Asiatic, Austroasiatic, and Sino-Tibetan language families exhibit the greatest standard deviations in their results. Figure 7 shows the performance results across various language scripts on the SIB-200 dataset for different values of k . It is evident that the Latin script generally achieves the highest performance, albeit with the highest standard deviation. Conversely, the scripts Nkoo, Olck, Tibt, and Tfng exhibit the lowest performance.

Parameter	NusaX	SIB-200	MASSIVE	LinCE SA	NollySenti	FIRE 2020
batch size	32	8	32	16	16	16
learning rate	1e-5	1e-5	1e-5	5e-5	5e-5	1e-5
max epoch	100	100	100	20	20	100
early stopping	3	5	3	5	5	5
adam beta 1	0.9	0.9	0.9	0.9	0.9	0.9
adam beta 2	0.999	0.999	0.999	0.999	0.999	0.999
adam epsilon	1e-8	1e-8	1e-8	1e-8	1e-8	1e-8

Table 9: Hyper-parameters for fine-tuning baselines.

Parameter	HF models	APIs
top-p	1	1
seed	-	42
temperature	0.2	0
max new tokens	10	64

Table 10: Hyper-parameters for model inference using Hugging Face models, such as BLOOMZ, mT0, XGLM, Aya-23, Aya-101, Gemma 1.1 7B Instruct, and Llama 8B Instruct and APIs, including Command-R, GPT-3.5 Turbo and GPT-4o.

C.3 ICL Classification Results

Table 16 presents the complete results for ICL classification task in Mono, XS, CS, and XS CS settings.

D Prompt Examples

Prompt examples used for ICL classification are provided in Tables 17 and 18. Specifically, we use two different templates: for direct prediction, label options are added to the prompt; for prediction by calculating label probabilities, label options are omitted, resulting in shorter prompts.

E DistFuse

We conduct a simplified hyper-parameter tuning process to determine the optimal weights for each model. Due to time constraints, we explore only a few weight combinations. For DistFuse (2), we evaluate two combinations: (1) $[\alpha = 1 \text{ and } \beta = 1]$, and (2) $[\alpha = 1 \text{ and } \beta = 3]$. For Dist (3), we assess three combinations: (1) $[\alpha = 1, \beta = 1, \gamma = 1]$, (2) $[\alpha = 1, \beta = 1, \gamma = 3]$, and (3) $[\alpha = 1, \beta = 2, \gamma = 3]$.

E5 _{LARGE}			XLM-R _{BASE}		
sample	label	dist	sample	label	dist
Query: Cepak saka hotelku nginep, namung digawa mlaku, ing kene akeh tenan pilihan panganane, panggonane sing amba, lan nyenengake Translation (in English): Near the hotel I stayed in, reachable by foot, so many food choice here, the place is huge, and fun Label: positive					
Miturutku mangan ana ing kene porsine akeh lan regane murah, banjur panganane cepet tekane maneh lan panggonane uga resik lan amba Translation (in English): In my opinion, eating here will grant you large portions for a cheap price, add to the fact that it's served quickly, too, and the place being clean and wide.	positive	0.436	Panggonan iki nyediakake pirang-pirang panganan, nanging sing aku jajal mesthi wae batagore, panggonane . uga resik Translation (in English): This place served several food, but of course the one I tried was the batagor, . place was clean too	positive	0.923
Aku seneng banget mangan ning restoran iki, menu masakane rena-rena, rasane enak, regane. ora tek larang Translation (in English): I really love eating in this restaurant. Varied menu, awesome flavours, and not really that expensive.	positive	0.452	Timku bakal nganakake mangan mbengi tema ing burgundy. Katimbang ilang, aku lan carikku njajal ngecek panggonane ndhisik sadina sakdurunge. Sisan uga tes panganane. Dalane adoh banget lan munggah medun bukit. Luwih nemen maneh pas dhewe mara mrana kuwi dina minggu sore dadi macet. Tekane ing kana sih pemandangane oke. Nanging model restorane biasa wae. Translation (in English): My team will be having a theme dinner in burgundy. Instead of getting lost, my secretary and I tried to check the place first the day before. Also test the food. The journey is very long and goes up and down hills. What made it worse was that when we went there it was a Sunday afternoon so there was traffic jam. When we got there, the view was okay. But the restaurant layout is ordinary	negative	0.972
Ing restoran iki panganan kang disediakake akeh banget lan regane cukup kajangkau, kahanane sek enak lan nyaman Translation (in English): In this restaurant there is a lot of food provided and the prices are quite affordable, the atmosphere is delicious and comfortable	positive	0.452	Pithik gorenge enak ing kene. Cocok kanggo sing lebare perjalanan adoh. Aku marang kene mulih saka njaba kutha, dadi mangane pas ngelih ngono deh. Marakake weteng wareg, panganan enak banjur pelayanane mantap. Kasire ayu ayu Translation (in English): The fried chicken is amazing here. Perfect after a long trip. I came here after returning out of town, so I was absolutely starving. My stomach was filled right back up. The food was good and servers were great. Not to mention, the cashiers were beautiful	positive	0.974
Panggonan iki nyediakake pirang-pirang panganan, nanging sing aku jajal mesthi wae batagore, . panggonane uga resik Translation (in English): This place served several food, but of course the one I tried was the batagor, place was clean too.	positive	0.467	Bingung arep mangan nandi sing panggone asik, panganan enak lan regane terjangkau? Mrene ae. Aku lan bojoku nikmati banget. Sing mara akeh-akeh cah enom dadi melu-melu ngrasa enom maneh. Translation (in English): Don't know where to have a nice and affordable place to grab a bite? Just come right here. Me and my partner are really enjoying it. Most of the customers are young people, making us feel just as young again.	positive	1.009
Kuota dadi entek resik kanggo ndelok foto-foto sing mung gawe aku sreji, panganan enak-enak sing marai ngiler Translation (in English): My quota is drained dry just to see photos that make me jelly, and delicious food that makes my mouth water.	negative	0.477	Panganan lumayan, nanging ana pelayan sing lumayan kemprom war dadi kurang nyaman. Kanggo panganan rada cepet yo ben ora kangelihen konsumene. Isih akeh sing kudu ditingkatake. Translation (in English): The food was okay, but there was this one server who was kinda dirty, making it a little less comfortable. Please serve the food quicker so the customers won't get hungry. There are many things to improve.	negative	1.018

Table 11: Retrieved samples from E5_{LARGE} and XLM-R_{BASE}.

LaBSE			
sample	label	dist	
Query: Cepak saka hotelku nginep, namung digawa mlaku, ing kene akeh tenan pilian panganane, panggonane sing amba, lan nyenengake			
Translation (in English): Near the hotel I stayed in, reachable by foot, so many food choice here, the place is huge, and fun			
Label: positive			
Ing restoran iki panganan kang disediakake akeh banget lan regane cukup kajangkau, kahanane sek enak lan nyaman	positive	0.896	
Translation (in English): In this restaurant there is a lot of food provided and the prices are quite affordable, the atmosphere is delicious and comfortable			
Wektu pengen mangan variasi panganan, piliane mesthi Hanamasa. Lokasi panggonane cukup enak. Pilian panganane akeh, saka awit camilan, bakar-bakaran, godhokan nganti panganan panutup. Ora nggelakne banget.	positive	0.934	
Translation (in English): When you wanna enjoy a variety of food, the first choice has to be Hanamasa. The location's pretty great. Lots of food you can choose from, ranging from snacks, barbeques, boiled food, all the way to desserts. Not bad at all!			
Mangan abreng karo dulur-dulur wedok kala wingi, panggon nyaman, enak kanggo nongkrong, pelayanane apik. Wis ping bolak-balik mangan ning kene.	positive	0.938	
Translation (in English): Dined togetha with da sistahs a lil' bit ago, cosy place, nice to hang out, good service. Have gone ta this place multiple times.			
Aku seneng banget mangan ning restoran iki, menu masakane rena-rene, rasane enak, regane ora tek larang.	positive	0.940	
Translation (in English): I really love eating in this restaurant. Varied menu, awesome flavours, and not really that expensive.			
Pithik gorenge enak ing kene. Cocok kanggo sing lebare perjalanan adoh. Aku marang kene mulih saka njaba kutha, dadi mangane pas ngelih ngono deh. Marakake weteng wareg, panganane enak banjur pelayanane mantap. Kasire ayu ayu	positive	0.946	
Translation (in English): The fried chicken is amazing here. Perfect after a long trip. I came here after returning out of town, so I was absolutely starving. My stomach was filled right back up. The food was good and servers were great. Not to mention, the cashiers were beautiful			

Table 12: Retrieved samples from LaBSE.

Model	Cross-lingual (XL)						Code-Switching (CS)			Micro	Macro
	BUCC	NollySenti	NusaX	NusaT	Tatoeba	avg.	LinCE MT	PHINC	avg.	avg.	avg.
metric	F1	F1	F1	F1	F1		F1	F1			
Fine-tune (SOTA)	99.00	N/A	N/A	N/A	83.80	N/A	N/A	N/A	N/A	N/A	N/A
$k = 1$											
LaBSE	98.77	80.52	77.89	81.17	81.14	83.90	34.36	69.70	52.03	74.79	67.97
CMLM	98.64	58.06	55.64	63.08	78.43	70.77	29.34	55.91	42.62	62.73	56.70
E5 _{BASE}	98.33	63.40	68.01	63.52	68.06	72.26	29.17	57.41	43.29	63.99	57.78
E5 _{LARGE}	98.66	67.50	72.67	67.20	75.73	76.35	34.32	65.63	49.97	68.82	63.16
MPNet _{BASE} V2	98.05	22.64	38.46	40.52	61.60	52.25	14.04	37.70	25.87	44.72	39.06
MiniLM _{L12-E384}	57.98	8.26	7.52	19.66	30.70	24.82	4.85	14.95	9.90	20.56	17.36
Glott-500	17.90	11.49	5.78	27.65	10.58	14.68	5.76	27.52	16.64	15.24	15.66
XLM-R _{BASE}	39.70	7.59	8.05	20.97	12.62	17.79	4.15	17.06	10.61	15.73	14.20
XLM-R _{LARGE}	26.51	5.53	5.03	18.60	6.57	12.45	2.20	9.88	6.04	10.62	9.25
Cohere-Embedv3	98.76	62.91	76.51	69.13	74.66	76.39	34.44	72.07	53.25	69.78	64.82
OpenAI-Embedv3-large	98.98	35.84	70.94	73.74	65.61	69.02	46.60	90.87	68.73	68.94	68.88
DistFuse (2)	98.90	80.29	80.96	80.28	83.19	84.72	37.97	74.97	56.47	76.65	70.60
DistFuse (3)	98.90	77.02	81.25	77.61	81.64	83.28	37.23	76.42	56.83	75.72	70.06
$k = 5$											
LaBSE	99.12	89.94	84.98	88.57	89.15	90.35	56.69	79.17	67.93	83.95	79.14
CMLM	99.15	71.54	65.91	74.48	87.38	79.69	50.96	66.91	58.94	73.76	69.32
E5 _{BASE}	99.05	77.58	79.47	74.68	80.42	82.24	51.48	69.79	60.64	76.07	71.44
E5 _{LARGE}	99.19	79.70	82.60	78.76	86.27	85.30	56.80	75.96	66.38	79.90	75.84
MPNet _{BASE} V2	99.01	29.95	50.24	51.55	70.38	60.23	26.26	47.28	36.77	53.52	48.50
MiniLM _{L12-E384}	72.82	14.72	14.07	29.01	45.39	35.20	10.47	20.33	15.40	29.54	25.30
Glott-500	32.26	20.92	14.39	38.51	21.28	25.47	11.30	35.22	23.26	24.84	24.37
XLM-R _{BASE}	57.23	15.00	15.74	29.42	20.87	27.65	9.08	22.27	15.68	24.23	21.67
XLM-R _{LARGE}	41.06	10.66	10.73	25.65	13.61	20.34	4.24	11.68	7.96	16.80	14.15
Cohere-Embedv3	99.29	76.19	85.08	80.72	84.98	85.25	57.28	82.05	69.66	80.80	77.46
OpenAI-Embedv3-large	<u>99.43</u>	43.39	79.37	83.05	76.77	76.40	74.92	<u>94.64</u>	<u>84.78</u>	78.80	80.59
DistFuse (2)	99.21	90.66	88.08	89.13	91.02	91.62	61.70	84.21	72.95	86.29	82.29
DistFuse (3)	99.26	87.65	88.24	87.36	90.07	90.52	61.27	85.22	73.25	85.58	81.89
$k = 10$											
LaBSE	99.17	<u>92.62</u>	87.67	90.21	91.02	92.14	61.54	82.15	71.84	86.34	81.99
CMLM	99.17	77.54	70.81	78.35	89.53	83.08	56.44	70.72	63.58	77.51	73.33
E5 _{BASE}	99.18	83.07	83.76	78.52	83.89	85.68	56.85	74.34	65.59	79.94	75.64
E5 _{LARGE}	99.31	83.78	86.19	82.35	88.80	88.09	61.6	79.57	70.58	83.09	79.34
MPNet _{BASE} V2	99.13	32.75	55.52	55.61	73.33	63.27	30.61	50.74	40.67	56.81	51.97
MiniLM _{L12-E384}	78.08	20.25	19.84	33.34	51.89	40.68	13.52	23.29	18.41	34.32	29.55
Glott-500	39.08	26.72	20.60	43.06	27.69	31.43	14.01	38.30	26.15	29.92	28.79
XLM-R _{BASE}	63.56	20.01	20.66	33.19	25.61	32.61	11.50	24.83	18.16	28.48	25.39
XLM-R _{LARGE}	47.24	14.03	14.34	28.31	17.79	24.34	5.11	12.70	8.90	19.93	16.62
Cohere-Embedv3	99.39	81.16	88.01	84.24	87.56	88.07	62.23	84.82	73.52	83.92	80.80
OpenAI-Embedv3-large	99.50	47.01	82.55	85.61	80.20	78.97	79.75	95.36	87.56	81.43	83.27
DistFuse (2)	99.29	93.21	90.48	91.12	92.85	93.39	66.20	86.75	76.47	88.56	84.93
DistFuse (3)	99.39	90.45	<u>90.40</u>	<u>89.83</u>	<u>91.87</u>	<u>92.39</u>	<u>65.94</u>	87.54	76.74	<u>87.92</u>	<u>84.57</u>

Table 13: Results on bitext retrieval. **Bold** and underlined numbers present the best and second-best models.

Model	Monolingual (Mono)					Code-Switching (CS)			Micro	Macro
	MASSIVE	NollySenti	NusaX	SIB-200	avg.	FIRE 2020	LinCE SA	avg.	avg.	avg.
metric	Acc.	Acc.	F1	Acc.		Acc.	Acc.			
Random	1.67	33.33	33.33	14.29	20.66	25.00	33.33	29.00	23.49	24.83
Majority	7.03	50.00	18.44	25.00	25.12	53.90	55.78	54.84	35.03	39.98
Fine-tune (SOTA)	86.10	<u>88.80</u>	80.00	75.90	82.70	N/A [‡]	N/A [‡]	N/A	N/A	N/A
Fine-tune (XLM-R _{BASE})	<u>85.04</u>	<u>87.16</u>	<u>75.43</u>	70.55	<u>79.55</u>	68.78	55.78	<u>62.28</u>	73.79	70.92
$k = 1$										
LaBSE	76.55	80.04	62.23	61.14	69.99	56.56	49.92	53.24	64.41	61.62
CMLM	76.24	79.48	63.40	60.42	69.89	54.83	48.63	51.73	63.83	60.81
E5 _{BASE}	74.82	82.96	65.59	62.23	71.40	57.14	50.03	53.59	65.46	62.49
E5 _{LARGE}	76.67	85.24	67.14	66.64	73.92	58.25	51.00	54.63	67.49	64.27
MPNet _{BASE} V2	69.41	75.24	53.29	56.24	63.55	51.21	49.70	50.46	59.18	57.00
MiniLM _{L12-E384}	63.32	72.28	58.35	39.77	58.43	51.49	49.00	50.25	55.70	54.34
Glott-500	64.01	75.52	57.00	51.76	62.07	53.48	48.84	51.16	58.44	56.62
XLM-R _{BASE}	61.93	74.56	58.29	43.66	59.61	53.57	47.44	50.51	56.57	55.06
XLM-R _{LARGE}	60.39	73.36	57.62	40.66	58.01	52.17	47.18	49.68	55.23	53.84
Cohere-Embedv3	77.78	86.80	68.54	71.08	76.05	59.30	51.43	55.37	69.16	65.71
OpenAI-Embedv3-large	74.97	79.56	63.61	67.44	71.40	61.19	51.37	56.28	66.36	63.84
DistFuse (2)	78.18	84.72	66.65	68.32	74.47	58.89	50.73	54.81	67.92	64.64
DistFuse (3)	78.59	86.24	67.44	70.76	75.76	59.15	50.94	55.05	68.85	65.40
$k = 5$										
LaBSE	78.62	82.08	66.90	64.67	73.07	63.65	53.85	58.75	68.30	65.91
CMLM	78.38	80.60	67.07	64.62	72.67	61.73	54.87	58.30	67.88	65.48
E5 _{BASE}	77.13	85.96	69.16	66.82	74.77	63.38	55.51	59.45	69.66	67.11
E5 _{LARGE}	79.10	87.20	71.72	71.05	77.27	64.14	57.40	60.77	71.77	69.02
MPNet _{BASE} V2	71.24	79.12	54.76	59.20	66.08	56.48	54.76	55.62	62.59	60.85
MiniLM _{L12-E384}	65.16	76.28	63.84	44.56	62.46	57.96	52.23	55.10	60.01	58.78
Glott-500	65.72	78.60	60.08	57.49	65.47	59.65	51.37	55.51	62.15	60.49
XLM-R _{BASE}	63.54	76.24	61.32	48.22	62.33	60.35	53.42	56.89	60.52	59.61
XLM-R _{LARGE}	62.08	76.20	60.57	45.44	61.07	59.11	52.56	55.84	59.33	58.45
Cohere-Embedv3	80.15	88.12	71.00	74.73	78.50	65.12	57.56	61.34	72.78	69.92
OpenAI-Embedv3-large	77.32	80.64	67.77	69.88	73.90	66.19	56.27	61.23	69.68	67.57
DistFuse (2)	80.42	87.00	71.90	72.13	77.86	64.21	56.16	60.19	71.97	69.02
DistFuse (3)	80.92	88.48	71.70	74.63	78.93	64.69	57.13	60.91	72.93	69.92
$k = 10$										
LaBSE	78.47	82.48	67.39	65.50	73.46	64.73	56.54	60.64	69.19	67.05
CMLM	78.21	82.04	67.11	64.84	73.05	62.96	55.57	59.27	68.46	66.16
E5 _{BASE}	77.18	86.36	69.07	67.72	75.08	64.71	57.61	61.16	70.44	68.12
E5 _{LARGE}	79.02	88.00	71.15	71.91	77.52	65.30	58.53	61.92	72.32	69.72
MPNet _{BASE} V2	70.75	80.40	53.85	59.67	66.17	59.26	57.40	58.33	63.56	62.25
MiniLM _{L12-E384}	64.47	77.12	64.27	46.87	63.18	60.61	53.95	57.28	61.22	60.23
Glott-500	65.14	79.36	58.69	59.47	65.67	62.04	54.17	58.11	63.15	61.89
XLM-R _{BASE}	62.98	78.40	62.72	50.39	63.62	62.06	54.44	58.25	61.83	60.94
XLM-R _{LARGE}	61.58	77.56	60.62	47.29	61.76	60.92	53.68	57.30	60.28	59.53
Cohere-Embedv3	80.15	88.64	69.87	<u>75.57</u>	78.56	65.88	58.36	62.12	73.08	70.34
OpenAI-Embedv3-large	77.27	82.28	66.80	<u>69.54</u>	73.97	<u>67.33</u>	58.20	62.77	70.24	68.37
DistFuse (2)	80.38	88.28	71.83	72.88	78.34	65.73	58.53	62.13	72.94	70.24
DistFuse (3)	80.79	88.96	70.99	75.32	79.02	65.97	<u>58.42</u>	62.20	<u>73.41</u>	<u>70.61</u>

Table 14: Results on retrieval-based classification. **Bold** and underlined numbers present the best and second-best models. [‡]For FIRE 2020, we modify the labels, thus there are no comparable results in the literature. For LinCE SA, we evaluate on the development split and we could not find any comparable result in the literature.

Model	Cross-lingual (XL)					Code-Switching (CS)	Micro	Macro
	MASSIVE	NollySenti	NusaX	SIB-200	avg.	FIRE 2020	avg.	avg.
source lang. metric	eng Acc.	en Acc.	eng F1	eng_Latn Acc.		tamil Acc.		
Random	1.67	33.33	33.33	14.29	20.66	25.00	21.52	21.09
Majority	7.03	50.00	18.44	25.00	25.12	41.91	28.48	26.80
Fine-tune (SOTA)	70.60	N/A	52.08	<u>69.10</u>	N/A	N/A [†]	N/A	N/A
Fine-tune (XLM-R _{BASE})	68.94	74.95	56.71	63.10	65.92	34.64	59.67	62.79
$k = 1$								
LaBSE	73.96	79.80	63.65	60.18	69.40	32.94	62.11	65.75
CMLM	73.08	74.00	58.98	57.51	65.89	34.87	59.69	62.79
E5 _{BASE}	63.43	74.30	34.08	63.11	58.73	35.53	54.09	56.41
E5 _{LARGE}	69.38	79.85	40.73	67.63	64.40	35.91	58.70	61.55
MPNet _{BASE} v2	46.05	61.60	48.44	55.95	53.01	32.12	48.83	50.92
MiniLM _{L12-E384}	35.72	62.20	41.15	30.50	42.39	31.60	40.23	41.31
Glott-500	24.66	66.70	44.45	40.08	43.97	33.16	41.81	42.89
XLM-R _{BASE}	27.49	64.85	36.41	33.98	40.68	32.42	39.03	39.86
XLM-R _{LARGE}	20.38	66.50	34.19	28.04	37.28	31.75	36.17	36.72
Cohere-Embedv3	70.87	<u>81.30</u>	65.29	69.67	71.78	35.68	64.56	68.17
OpenAI-Embedv3-large	61.09	67.85	65.45	67.36	65.44	31.90	58.73	62.08
$k = 5$								
LaBSE	75.80	81.80	68.25	63.75	72.40	38.58	65.64	69.02
CMLM	75.48	78.70	64.89	58.89	69.49	38.72	63.34	66.41
E5 _{BASE}	66.83	73.45	51.82	67.43	64.88	40.28	59.96	62.42
E5 _{LARGE}	72.48	78.60	60.99	71.53	70.90	40.28	64.78	67.84
MPNet _{BASE} v2	50.83	64.00	53.98	58.73	56.89	38.58	53.22	55.05
MiniLM _{L12-E384}	40.19	65.55	52.79	34.81	48.34	36.80	46.03	47.18
Glott-500	28.67	73.50	49.37	47.01	49.64	37.61	47.23	48.43
XLM-R _{BASE}	31.27	69.15	39.52	39.89	44.96	38.58	43.68	44.32
XLM-R _{LARGE}	24.74	69.20	36.13	34.19	41.07	37.69	40.39	40.73
Cohere-Embedv3	74.18	78.60	64.59	74.62	73.00	40.28	<u>66.45</u>	69.73
OpenAI-Embedv3-large	63.62	66.15	<u>69.22</u>	69.09	67.02	38.43	61.30	64.16
DistFuse (2)	77.53	79.25	63.74	65.03	71.39	39.24	64.96	68.17
DistFuse (3)	77.27	78.25	61.67	66.00	70.80	38.65	64.37	67.58
$k = 10$								
LaBSE	75.89	81.20	68.54	65.29	72.73	41.10	66.40	69.57
CMLM	75.77	<u>81.30</u>	66.06	58.11	70.31	40.88	64.42	67.37
E5 _{BASE}	67.60	74.20	51.54	68.71	65.51	42.73	60.96	63.23
E5 _{LARGE}	73.09	77.50	61.40	72.33	71.08	41.99	65.26	68.17
MPNet _{BASE} v2	56.45	64.80	57.88	59.61	59.69	41.25	56.00	57.84
MiniLM _{L12-E384}	42.07	66.55	58.66	37.34	51.16	39.61	48.85	50.00
Glott-500	30.73	74.10	51.50	50.67	51.75	40.06	49.41	50.58
XLM-R _{BASE}	32.96	70.45	45.11	41.83	47.59	41.02	46.27	46.93
XLM-R _{LARGE}	27.18	69.50	39.62	39.20	43.88	39.47	42.99	43.43
Cohere-Embedv3	74.98	77.95	61.69	76.06	<u>72.67</u>	<u>42.36</u>	66.61	<u>69.64</u>
OpenAI-Embedv3-large	64.43	65.15	69.88	69.07	67.13	40.50	61.81	64.47
DistFuse (2)	77.75	78.30	62.72	64.71	70.87	40.73	64.84	67.86
DistFuse (3)	<u>77.67</u>	76.70	58.94	67.43	70.19	41.77	64.50	67.34

Table 15: Results on retrieval-based classification in the cross-lingual setting. The source language is English for all datasets except FIRE 2020, where the source language is Tamil. **Bold** and underlined numbers present the best and second-best models. [†]We preprocess the dataset differently from the original dataset. Thus, there are no comparable results in the literature.

Model	Mono				XL				CS			XL CS	Micro	Macro
	NollySenti	NusaX	SIB-200	avg.	NollySenti	NusaX	SIB-200	avg.	FIRE 2020	LinCE SA	avg.	FIRE 2020	avg.	avg.
metric	Acc.	F1	Acc.		Acc.	F1	Acc.		Acc.	Acc.		Acc.		
BLOOMZ 560m														
$k = 0$	70.68	29.01	37.94	45.88	65.40	37.87	26.82	43.36	16.25	55.41	35.83	12.09	39.05	34.29
$k = 1$														
LaBSE	80.20	62.79	60.19	67.73	81.60	63.57	58.54	67.90	55.82	51.10	53.46	33.61	60.82	55.68
E5 _{LARGE}	82.64	66.94	67.54	72.37	82.00	66.41	67.54	71.98	57.94	50.56	54.25	36.35	64.21	58.74
Cohere-Embedv3	83.40	66.44	69.43	73.09	82.05	65.02	67.70	71.59	58.12	52.18	55.15	35.98	64.48	58.95
BLOOMZ 1.7B														
$k = 0$	82.28	47.03	33.00	54.10	79.25	46.34	33.00	52.86	17.55	53.85	35.70	11.80	44.90	38.62
$k = 1$														
LaBSE	84.60	54.27	62.00	66.96	81.75	55.81	60.50	66.02	57.19	56.75	56.97	35.39	60.92	56.33
E5 _{LARGE}	86.48	58.14	69.51	71.38	82.50	60.16	69.28	70.65	<u>59.05</u>	57.02	<u>58.04</u>	38.50	64.52	59.64
Cohere-Embedv3	86.48	58.80	71.31	72.20	82.50	57.48	69.36	69.78	59.27	57.07	58.17	37.69	64.44	59.46
BLOOMZ 3B														
$k = 0$	79.48	45.99	34.12	53.20	76.25	45.07	34.02	51.78	14.16	58.47	36.32	9.50	44.12	37.70
$k = 1$														
LaBSE	85.68	64.98	62.37	71.01	82.95	62.08	61.65	68.89	57.99	55.14	56.57	37.46	63.37	58.48
E5 _{LARGE}	<u>86.52</u>	66.68	69.05	74.08	<u>83.70</u>	65.69	70.17	73.19	59.41	55.46	57.44	39.09	66.20	60.95
Cohere-Embedv3	86.88	66.80	70.59	74.76	82.40	61.05	69.72	71.06	59.19	56.11	57.65	38.58	65.70	60.51
mT0 3B														
$k = 0$	83.96	28.18	47.74	53.29	83.35	30.09	47.48	53.64	54.18	26.04	40.11	42.51	49.28	47.39
$k = 1$														
E5 _{LARGE}	85.12	39.34	52.60	59.02	81.55	36.87	55.16	57.86	54.17	39.16	46.67	42.36	54.04	51.48
XGLM 564m														
$k = 0$	60.80	32.11	24.84	39.25	55.50	31.24	24.83	37.19	11.91	47.93	29.92	10.46	33.29	29.21
$k = 1$														
E5 _{LARGE}	23.76	35.05	52.97	37.26	33.25	36.50	50.60	40.12	21.61	23.67	22.64	12.83	32.25	28.21
XGLM 2.9B														
$k = 0$	63.84	38.84	24.56	42.41	58.20	37.76	24.53	40.16	11.97	57.45	34.71	10.39	36.39	31.92
$k = 1$														
E5 _{LARGE}	39.72	32.56	55.43	42.57	52.60	36.60	57.09	48.76	14.61	40.29	27.45	10.39	37.70	32.29
Aya-23 8B														
$k = 0$	61.12	39.59	18.94	39.88	54.45	37.26	18.94	36.88	54.44	52.99	53.72	43.18	42.32	43.42
$k = 1$														
LaBSE	56.24	68.24	63.67	62.72	55.35	67.81	58.76	60.64	56.66	47.71	52.19	36.42	56.76	52.99
E5 _{LARGE}	54.52	67.57	68.90	63.66	56.15	67.54	66.89	63.53	58.85	47.39	53.12	38.50	58.48	54.70
Cohere-Embedv3	54.16	67.66	69.61	63.81	54.05	68.51	64.72	62.43	58.77	46.53	52.65	37.17	57.91	54.01
Aya-101 13B														
$k = 0$	84.40	77.78	73.78	78.65	82.35	76.98	73.83	77.72	35.25	49.33	42.29	26.26	64.44	56.23
$k = 1$														
E5 _{LARGE}	86.40	<u>79.19</u>	77.42	<u>81.00</u>	85.80	79.24	75.56	80.20	48.59	53.20	50.90	36.20	69.07	62.08
Gemma 1.1 7B Instruct														
$k = 0$	71.20	52.68	42.64	55.51	67.05	50.21	42.82	53.36	47.47	55.78	51.62	37.24	51.90	49.43
$k = 1$														
E5 _{LARGE}	76.00	56.20	65.26	65.82	74.85	52.90	65.71	64.49	48.14	58.10	53.12	35.68	59.20	54.78
Llama 3 8B Instruct														
$k = 0$	71.60	57.77	57.82	62.40	66.95	56.46	57.82	60.41	46.81	58.63	52.72	36.05	56.66	52.90
$k = 1$														
LaBSE	83.16	64.86	67.48	71.83	78.15	63.34	62.50	68.00	48.57	59.01	53.79	34.94	62.45	57.14
E5 _{LARGE}	85.04	66.59	72.92	74.85	77.65	64.34	66.83	69.61	49.82	58.42	54.12	35.68	64.14	58.57
Cohere-Embedv3	85.76	66.79	73.64	75.40	74.70	62.31	67.21	68.07	49.64	58.74	54.19	37.02	63.98	58.67
Llama 3.1 8B Instruct														
$k = 0$	74.88	49.85	57.04	60.59	70.85	48.66	57.07	58.86	37.45	58.53	47.99	26.56	53.43	48.50
$k = 1$														
E5 _{LARGE}	86.36	58.70	72.99	72.68	78.45	32.49	66.05	59.00	49.37	58.85	54.11	35.16	59.82	55.24
Command-R														
$k = 0$	65.16	35.27	43.50	47.98	59.25	35.42	43.39	46.02	50.72	58.96	54.84	44.44	48.46	48.32
$k = 1$														
E5 _{LARGE}	67.96	39.21	67.91	58.36	62.30	41.45	66.92	56.89	55.10	58.58	56.84	41.99	55.71	53.52
GPT-3.5 Turbo														
$k = 0$	68.80	63.96	68.53	67.10	63.30	63.64	68.46	65.13	50.65	57.99	54.32	45.18	61.17	57.93
$k = 1$														
LaBSE	77.12	62.53	71.43	70.36	75.25	65.65	72.03	70.98	53.58	60.95	57.27	42.14	64.52	60.19
E5 _{LARGE}	77.24	63.30	72.48	71.01	75.25	65.97	73.47	71.56	53.84	<u>60.41</u>	57.13	42.73	64.97	60.61
Cohere-Embedv3	77.16	63.07	72.23	70.82	74.20	66.52	73.27	71.33	52.90	60.14	56.52	41.84	64.59	60.13
GPT-4o														
$k = 0$	83.16	77.08	<u>79.53</u>	79.92	81.55	<u>76.42</u>	<u>79.47</u>	<u>79.15</u>	49.89	57.07	53.48	53.04	<u>70.80</u>	<u>66.40</u>
$k = 1$														
E5 _{LARGE}	85.04	79.52	82.15	82.24	83.20	78.96	80.69	80.95	57.25	57.02	57.14	<u>49.26</u>	72.57	67.40

Table 16: Results on ICL classification. **Bold** and underlined numbers present the best and second-best models.

Template	Instruction:<INSTRUCTION> Please only output the label. <FEW-SHOT SAMPLE> Options:<OPTIONS> Input:<QUERY> Prediction:
Few-shot sample	Input:<INPUT TEXT>. Prediction:<LABEL>
Dataset	Prompt
FIRE 2020	Instruction:Generate a sentiment label for a given input. Please only output the label. Input: Ikka waiting..... Prediction:Positive Options:['Positive', 'Negative', 'Mixed', 'Unknown'] Input:mind blowing ikkaaaa.... Prediction:
LinCE SA	Instruction:Generate a sentiment label for a given input. Please only output the label. Input:@brissamayen Thanks :) ay si todavia le hablas a mi chikiya in the future te invitamos a la boda :) lol 2665 Prediction:positive Options:['negative', 'neutral', 'positive'] Input:@brissamayen @sanluispotoyees estopp I blashhh lol jk but aww :) thanks haha (x Prediction:
NollySenti	Instruction:Generate a sentiment label for a given input. Please only output the label. Input:Enjoy! Very nice... very nice indeed. Prediction:positive Options:['negative', 'neutral', 'positive'] Input:Damn.....so interesting Prediction:
NusaX	Instruction:Generate a sentiment label for a given input. Please only output the label. Input:Kawanulun bagawi di gojek Prediction:neutral Options:['negative', 'neutral', 'positive'] Input:Macet di mana-mana pasl agi peraian Prediction:
SIB200	Instruction:Generate a topic label for a given input. Please only output the label. Input:Batu kabidi bateka mikalu bua njila ya makasa ni ya makalu. Prediction:travel Options:['geography', 'science/technology', 'entertainment', 'politics', 'health', 'travel', 'sports'] Input:Anu kaniemesha uvua mutapika bibi ku mutu. Prediction:

Table 17: Prompt examples. $k = 1$ with LaBSE.

Template	Instruction:<INSTRUCTION> Please only output the label. <FEW-SHOT SAMPLE> Options:<OPTIONS> Input:<QUERY> Prediction:
Few-shot sample	Input:<INPUT TEXT>. Prediction:<LABEL>
Dataset	Prompt
FIRE 2020	Instruction:Generate a sentiment label for a given input. Please only output the label. Input: Njan mathram aano sunny chechiye kaanan vannath Sunny chechi uyir Prediction:Positive Options:['Positive', 'Negative', 'Mixed', 'Unknown'] Input:Sunny chechiye kaanan vannathu njan maathram aano Prediction:
LinCE SA	Instruction:Generate a sentiment label for a given input. Please only output the label. Input:hablar de los planes de spring break y mis 18 me pone bien hyper ! :D Prediction:positive Options:['negative', 'neutral', 'positive'] Input: Prediction:
NollySenti	Instruction:Generate a sentiment label for a given input. Please only output the label. Input:Amazing Film. . . . Indeed the most anticipated film from Nollywood 2019 didn't disappoint. Loved it all. Well done to Genevieve and Team. Prediction:positive Options:['negative', 'neutral', 'positive'] Input:This is the nollywood evolution. . . . This is arguably my best Nigeria movie for year 2019. I cannot find any misplaced in this movie, perfectly executed, simple and so informative about our society n thought provoking on career part for our children Prediction:
NusaX	Instruction:Generate a sentiment label for a given input. Please only output the label. Input:Tempatnya nyaman banget, makanannya enak, kopinya enak. Pas buat nongkrong bareng teman-teman atau makan malam. Prediction:positive Options:['negative', 'neutral', 'positive'] Input:Tempat yang bagus kalau dinikmati malam hari. Cukup nyaman. Harga cukup terjangkau. Favorit saya steak tenderloinnya. Cukup enak. Prediction:
SIB200	Instruction:Generate a topic label for a given input. Please only output the label. Kel sirvisu ta uzadu txeu pa transporti, inkluidu artizanatu di lazer, y tanb00ea ispidisonz ki ten nisisidadi di dadus y v00f3s a dist00e1nsia. Prediction:science/technology Options:['geography', 'science/technology', 'entertainment', 'politics', 'health', 'travel', 'sports'] Input:Sist00e9ma di IA gosi ta uzadu kuazi txeu na 00e1rias di ikonumia, midisina, injinharia y militar, sima ten stadu ta podu na txeu komputer di kaza y software di v00eddio geimi. Prediction:

Table 18: Prompt examples. $k = 1$ with E5_{LARGE}.